

NullState: a reference-anchored framework for detecting, diagnosing, and quantifying the off-target compartment of human neural organoids

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Abstract

Human neural organoids reproducibly generate cell types that lie outside the developmental trajectories they are intended to model, but the field lacks a principled, atlas-anchored way to *quantify* this off-target compartment and ask whether it carries reproducible structure. We present **NullState**, a reference-based pipeline that maps the 1,767,346-cell Human Neural Organoid Atlas (HNOCA) onto a harmonized 2,615,898-cell first-trimester human fetal reference (HD-BCA + a fetal-body atlas) by variational reference surgery, assigns each organoid cell a pair of calibrated mapping scores (off-manifold distance and label entropy), and classifies it into a six-state off-target taxonomy. Across the atlas, 38.4% of cells fall into an *ambiguous-novel* (off-manifold) compartment. Four orthogonal diagnostics show this compartment is not a projection artifact but a *structured mixture* of a reference-coverage gap and a pervasive in-vitro stress program: off-manifold cells lose neural-progenitor identity while gaining hypoxia and unfolded-protein-response signatures, their burden rises monotonically with maturation and metabolic load, varies ~ 240 -fold across differentiation protocols (0.4% to 96.8%), and concentrates into coherent transcriptomic clusters. We then show, as a reproducible *negative* result, that a shared-latent contrastive generative model cannot separate lineage from in-vitro state here: an origin-decoding adversary recovers organoid-versus-primary identity from the putatively dish-stripped latent at ≥ 0.998 balanced accuracy, robustly across latent-capacity and batch-conditioning interventions, because the in-vitro state is transcriptome-pervasive rather than a separable nuisance subspace. Finally, we demonstrate that off-target geometry is nonetheless recoverable: among well-powered off-target populations, all pairwise lineage distances are real (clearing a matched- N noise floor by 14–105 $\times$) and biologically ordered, with the two myogenic populations closest in the embedding. NullState provides a deployable off-target quality axis for organoid studies and a precise account of when latent disentanglement is and is not attainable.

1 Introduction

Human pluripotent-stem-cell-derived neural organoids have become a central model of human brain development [1, 2], capturing cortical cytoarchitecture, diverse neuronal and glial identities, and functional network activity [3–5]. Yet a recurring concern limits their interpretation: organoids also produce cells that do not correspond to any *bona fide* in-vivo neural state. These include non-neural lineages arising from incomplete germ-layer restriction and aberrant states driven by the culture environment itself. A landmark observation is that metabolic and endoplasmic-reticulum (ER) stress

in organoids impairs the specification of correct molecular subtypes [6], and cross-protocol comparisons against fetal tissue have repeatedly surfaced protocol-dependent, off-target differentiation routes [7, 8]. The recent Human Neural Organoid Atlas (HNOCA)—an integrated transcriptomic atlas of >1.7 million organoid cells spanning dozens of protocols [9]—makes it possible, for the first time, to study this off-target compartment at scale and against a common coordinate system.

What has been missing is a *principled, reference-anchored* way to (i) decide which organoid cells are off-target, (ii) explain *why* they are off-target, and (iii) ask whether the off-target compartment is merely noise or carries reproducible biological structure. Reference-based mapping provides the natural substrate: by projecting organoid cells onto a comprehensive fetal atlas, cells that map confidently to a fetal identity can be separated from those that do not. Two recently completed human references make such anchoring feasible—a comprehensive first-trimester human brain cell atlas (HDBCA) [10] and a human fetal-body gene-expression atlas [11]—together spanning neural and non-neural developmental origins.

Here we introduce **NullState**, a pipeline that operationalizes this idea (Fig. 1). We harmonize the two fetal atlases into a single 2,615,898-cell reference, map the 1,767,346-cell HNOCA query onto it by variational reference surgery [12–14], and assign every organoid cell two calibrated mapping scores—an off-manifold reconstruction distance and a label-assignment entropy—that jointly define a six-state off-target taxonomy. We organize the analysis around three questions: **(Aim 1)** is the off-target measurement well-posed and adequately powered? **(Aim 2)** can lineage biology be disentangled from the in-vitro state, so that off-target cells can be placed on a “dish-stripped” lineage axis? and **(Aim 3)** do off-target populations occupy distinct, reproducible geometry? Our central findings are that the dominant off-manifold compartment is a structured biological mixture rather than an artifact; that shared-latent disentanglement provably fails here for an interpretable reason; and that off-target geometry is nevertheless recoverable and biologically coherent.

2 Results

2.1 A harmonized fetal reference supports confident organoid mapping

We assembled a fetal reference by combining the first-trimester HDBCA [10] with the non-neural compartments of a human fetal-body atlas [11], retrieving raw unique-molecular-identifier (UMI) counts from the CZ CELLxGENE Discover Census [15]. The merged reference comprises 2,615,898 cells $\times$ 61,497 genes, with 32/33 lineage origins resolved. We trained a single-cell variational model (scVI) for 400 epochs (evidence lower bound, ELBO, 695.6), refined it into a label-aware model (scANVI, ELBO 736.1), and projected the HNOCA query onto the reference by architectural surgery (scArches; query-surgery ELBO 920.2) [12–14, 16]. Surgery required two stabilizers that we found essential at atlas scale: excluding 9,229 non-UMI (full-length read-count) cells whose library sizes destabilize the negative-binomial encoder, and filtering to cells with ≥ 50 counts across highly variable genes (1,767,346 $\rightarrow$ 1,689,762; 4.4% removed).

Donor-level batch integration improved cross-batch mixing while preserving biological structure: the integration local inverse Simpson’s index (iLISI) rose from 6.10 (unintegrated principal components) to 8.10 after scVI, with only a modest change in the cell-type cLISI (1.31 $\rightarrow$ 1.44) [17, 18] (Fig. 2c). We calibrated two decision thresholds on held-out reference cells—a label-entropy threshold $\tau_H = 0.96$ and an off-manifold threshold $\tau_R = 1.93$ —and used them to assign each organoid cell to one of six states (Fig. 2a,b). The resulting partition is dominated by an *ambiguous-novel* (off-manifold) compartment (677,783 cells; 38.4%), followed by clean on-target (459,441; 26.0%), poorly differentiated on-target (281,217; 15.9%), ambiguous (167,838; 9.5%), true off-target (93,795; 5.3%), and quality-control-dropped cells (86,813; 4.9%).

Three feasibility gates confirmed the measurement is well-posed (Fig. 2d). The off-target taxonomy is internally consistent (Gate A, pass); an in-vitro “dish-vector” computed on clean on-target cells is highly self-consistent (Gate B, mean cosine 0.855 > 0.70), justifying a single shared in-vitro axis; and the true off-target compartment is adequately powered for downstream geometry (Gate C, green; 93,795 cells). NullState thus yields a calibrated, adequately powered off-target measurement over the entire organoid atlas.

2.2 The off-manifold compartment is a structured mixture of reference gap and in-vitro stress

The size of the ambiguous-novel compartment (38.4%) raises an essential question: is it a mapping artifact, or does it reflect real biology? Four orthogonal diagnostics, computed on the neural subset, converge on the latter (Fig. 3).

Identity and stress programs (Fig. 3a). Relative to clean on-target neural cells, ambiguous-novel neural cells show markedly reduced neural-progenitor identity (NPC score 1.11 $\rightarrow$ 0.61) while gaining canonical stress signatures: hypoxia (0.60 $\rightarrow$ 0.96) [19] and the unfolded-protein/ER-stress response (0.12 $\rightarrow$ 0.56, a $\sim$ 5-fold increase). An independent, atlas-precomputed glycolysis hallmark score is correspondingly elevated (0.46 $\rightarrow$ 0.65). This is the transcriptional signature of culture stress described by Bhaduri *et al.* [6], here resolved at atlas scale and tied directly to off-manifold status.

Maturation and metabolic load (Fig. 3b). The off-manifold fraction rises monotonically with a per-cell potency/complexity proxy (number of detected genes; quintile fractions 0.13 $\rightarrow$ 0.78) and is highest in the oldest organoids (age quintile fraction 0.62; median off-manifold score 1.84 $\rightarrow$ 2.12 across age) [20]. Off-target burden therefore accrues with culture time and metabolic demand, consistent with a stress-driven mechanism rather than random misassignment.

Protocol dependence (Fig. 3c). Across 26 differentiation protocols, the ambiguous-novel fraction spans nearly the entire range (0.4% to 96.8%; standard deviation 0.28). Guided cortical protocols sit at the low end (e.g., DAPT-patterned cortico-motor [21], 0.4%; 2, 10%), whereas regionally divergent or barrier-forming protocols dominate the high end—choroid-plexus organoids [22] reach 96.8%, midbrain-dopaminergic [23] and thalamic [24] protocols 85–90%. The off-manifold signal thus partly tracks *reference coverage*: the first-trimester, cortex-rich HDBCA under-represents later and non-cortical neural states, which are then read as off-manifold.

Cluster coherence (Fig. 3d,e). Ambiguous-novel cells are not scattered. Across 30 Leiden communities in the atlas embedding [25], the off-manifold fraction is bimodal (per-cluster s.d. 0.29 about an overall mean of 0.55); 20% of cells reside in near-pure off-manifold clusters ($\geq 80\%$) and 21% in near-pure clean clusters ($\leq 20\%$). Off-manifold status is a coherent population-level property, visible as contiguous territory in the scPoli UMAP [26].

Together these diagnostics establish that the dominant off-target compartment is a reproducible biological mixture: a *reference-coverage gap* (non-cortical and later neural states absent from a first-trimester reference) superimposed on a pervasive *in-vitro stress* program (hypoxia, unfolded-protein response, glycolysis) that intensifies with maturation.

2.3 Lineage cannot be disentangled from in-vitro state in a shared latent (a characterized negative)

If the in-vitro state were a low-dimensional nuisance, one could subtract it and place off-target cells on a clean lineage axis. We tested this directly with contrastiveVI, which factorizes a background (shared) latent z_{lin} from a salient latent z_{iv} [27], training the background on primary fetal cells and the salient on organoid cells, so that z_{lin} should encode dish-stripped lineage. We evaluated

disentanglement with a held-out adversary: a classifier that attempts to recover organoid-versus-primary origin from z_{lin} (balanced accuracy 0.5 is ideal; < 0.6 would indicate success) (Fig. 4a).

Disentanglement failed, and did so robustly. The baseline model ($\dim z_{\text{iv}} = 10$) yielded adversary balanced accuracy 0.998; tripling salient capacity ($\dim z_{\text{iv}} = 30$) gave 0.999; and adding a source batch covariate to the decoder—the textbook remedy for a nuisance axis confounded with a condition—gave 0.998. Training converged normally in every case (Fig. 4b), so this is not an optimization failure. The mechanism is structural (Fig. 4c): because the in-vitro state is transcriptome-pervasive (Fig. 3) and organoid and fetal cells occupy near-disjoint regions of expression space, the background *encoder* $q(z_{\text{lin}} | x)$ observes and encodes source identity directly; conditioning the *decoder* on a batch label cannot undo what the encoder has already written into z_{lin} . We therefore report, as a precise negative result, that shared-latent contrastive disentanglement is not attainable in the fetal-versus-organoid regime: the dish signature is not a separable subspace but a global transcriptional shift. This delimits where “calibration without anchoring” can and cannot be applied and motivates encoder-side or distributional alternatives.

2.4 Off-target populations occupy distinct, biologically coherent geometry

The disentanglement failure does not, however, invalidate the geometry of off-target populations relative to one another. Aim 3 compares off-target *organoid* populations against *each other*; because all such cells share the same organoid-versus-primary offset in z_{lin} , that offset is a constant translation that cancels exactly in pairwise sliced-Wasserstein distances ($W(\mathcal{A}+c, \mathcal{B}+c) = W(\mathcal{A}, \mathcal{B})$) [28]. The contamination shifts all populations together and therefore does not distort their relative structure.

We computed pairwise lineage distances among the true off-target populations, controlling for sample size by a rarefaction-derived minimum ($N_{\text{min}} = 2000$) and a matched- N self-distance noise floor; a pair is called only if its cross-distance confidence interval clears the floor (Fig. 5). Of the 17 off-target populations, four are well powered (cortical-adrenal 49,338; skeletal-muscle 30,629; fibroblast 6,349; cardiac-muscle 2,534). All six pairwise comparisons are real differences, each clearing the noise floor by 14 to $105\times$ (W from 0.32 to 0.99 against floors of 0.007–0.026); the remaining 130 small- N comparisons are correctly returned as underpowered. Crucially, the recovered geometry is *biologically ordered*: the two myogenic populations (cardiac and skeletal muscle) are closest ($W = 0.32$), fibroblast is most distant from the muscle populations ($W \approx 0.96$ –0.99), and the steroidogenic adrenal-cortical population sits between them, exactly as a lineage embedding—not a source-confounded one—should arrange them (Fig. 5a,d). Off-target geometry is thus both statistically real and lineage-interpretable.

3 Discussion

NullState turns the off-target problem of neural organoids into a quantitative, atlas-anchored measurement. Three results have practical consequences. First, the off-manifold compartment is large (38.4%) but *legible*: it decomposes into a reference-coverage gap and an in-vitro stress program, each separately actionable. The reference-gap component argues for extending fetal references beyond the first trimester and beyond cortex before off-manifold scores are interpreted as “aberrant” for non-cortical protocols; the stress component, intensifying with age and metabolic load and mirroring Bhaduri et al. [6], provides a per-cell, deployable readout for evaluating stress-mitigation strategies. The ~ 240 -fold protocol spread additionally yields a concrete, reference-anchored fidelity ranking across organoid platforms.

Second, our negative disentanglement result is, we believe, useful precisely because it is characterized. The widespread intuition that a contrastive or batch-aware latent can “subtract the dish” presupposes that the in-vitro state is a separable nuisance subspace. We show that in the fetal-versus-organoid regime it is not—it is a global, encoder-visible shift—so shared-latent methods cannot produce a source-invariant lineage axis regardless of salient capacity or decoder batch conditioning. Methods that intervene at the encoder (e.g., adversarial or invariance-penalized encoders) or that operate distributionally rather than per-cell are the appropriate next step.

Third, and reassuringly, the quantity Aim 3 actually needs—the *relative* geometry of off-target populations—survives the disentanglement failure by an exact cancellation argument, and recovers biologically sensible lineage structure. This means off-target geometry can be used as a quality and comparison axis today, even though a fully dish-stripped per-cell coordinate is not yet attainable.

Limitations follow directly from the analysis. Off-manifold status conflates reference gap with biological aberrance, so absolute off-target rates are reference-dependent and should be read comparatively. The dish-vector and stress scores are signature-based proxies. The well-powered Aim 3 populations are non-neural off-target lineages; neural off-target geometry awaits a reference that closes the coverage gap. Finally, pairwise distances are robust to the global source offset but may still include a component of *differential* in-vitro state between populations; the lineage-consistent ordering (myogenic populations closest) indicates that lineage dominates, but isolating that component is future work.

4 Methods

Reference construction. Raw-count fetal data were retrieved from the CZ CELLxGENE Discover Census [15]: the full first-trimester HDBCA [10] as the neural core, and non-neural compartments of a fetal-body atlas [11] filtered at the Census level to the relevant ontology cell types (mesoderm, endoderm, neural-crest origins). Gene identifiers were harmonized to Ensembl; counts were verified as integer UMIs prior to training. A lineage *origin* label was assigned to each reference cell through a curated cell-type→origin map; only the generic “cell” type was permitted to remain unmapped.

Mapping by reference surgery. We trained scVI (400 epochs, donor batch key) and scANVI on the reference, then mapped the HNOCA query [9] by scArches architectural surgery, using the implementations in scvi-tools [12–14, 16]. Organoid raw UMI counts were taken from the CELLxGENE HNOCA copy. Surgery stability at atlas scale required (i) dropping non-UMI read-count cells (per-cell maximum count > 1000) and (ii) a learning-rate, gradient-clipping, and KL-warmup schedule; without these the negative-binomial encoder diverged to NaN.

Scoring and classification. For each query cell we computed a label-assignment entropy H over scANVI posteriors and an off-manifold reconstruction distance ρ in the latent space. Thresholds τ_H and τ_R were calibrated on a held-out 10% of reference cells (off-manifold nearest-neighbor distances fit excluding the held-out set to avoid self-neighbor deflation). The six-state taxonomy combines on/off-manifold status (ρ vs. τ_R), label confidence (H vs. τ_H), and the lineage origin of the assigned label.

Diagnostics. Gene-set scores (neural progenitor, neuron, glia, hypoxia [19], unfolded-protein response, glycolysis) were computed with `scanpy.tl.score_genes` on log-normalized expression. Maturation was assessed against organoid age and a detected-gene potency proxy [20]; protocol

stratification used the atlas `assay_differentiation` annotation; cluster coherence used Leiden communities [25] on the scPoli embedding.

Disentanglement. `contrastiveVI` [27] was trained with primary fetal on-target cells as background and organoid on-target cells as target (100,000 each, 2000 highly variable genes, raw-count layer). Disentanglement was scored by a five-fold cross-validated logistic-regression adversary predicting organoid-versus-primary origin from the background latent z_{lin} (balanced accuracy). Interventions: salient dimension $10 \rightarrow 30$, and a source batch covariate registered through `setup_anndata`.

Geometry. For each off-target population pair we estimated the sliced-Wasserstein distance [28] between background-latent embeddings, with a per-population minimum sample size from rarefaction ($N_{\text{min}} = 2000$) and a matched- N self-distance noise floor (95th percentile). A pair was called a real difference when its bootstrap cross-distance interval exceeded the floor; multiplicity was controlled by Benjamini–Hochberg [29]. Sliced-Wasserstein pairwise distances are invariant to a translation shared by both populations, which is what makes the comparison valid despite the residual source offset in z_{lin} .

Compute and code. Models were trained on a single NVIDIA H100 GPU. All quantitative panels are generated directly from pipeline artifacts (`docs/manuscript/make_figures.py`); numerical values are exported to `numbers.json`.

NullState: a reference-based pipeline for organoid off-target detection and geometry

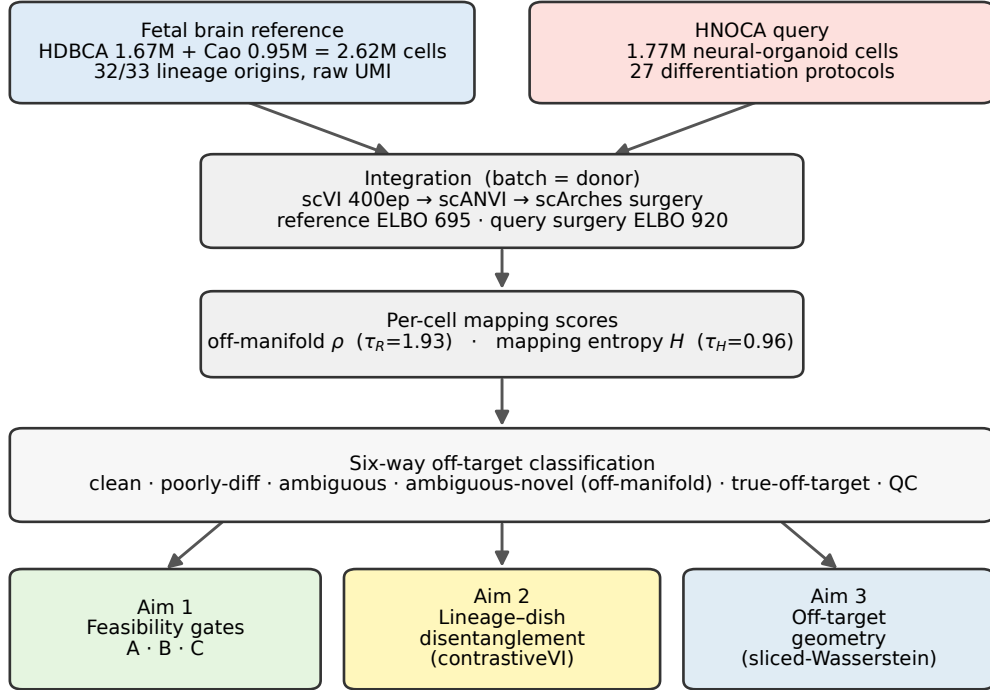


Figure 1. The NullState pipeline. A harmonized first-trimester fetal reference (HDBCA + fetal-body atlas; 2,615,898 cells, raw UMIs) and the HNOCA organoid query (1,767,346 cells, 27 protocols) are co-embedded by scVI/scANVI followed by scArches reference surgery. Each organoid cell receives two calibrated mapping scores—off-manifold distance ρ ($\tau_R = 1.93$) and label entropy H ($\tau_H = 0.96$)—that define a six-state off-target taxonomy, which feeds three analyses: feasibility gates (Aim 1), lineage–dish disentanglement (Aim 2), and off-target geometry (Aim 3).

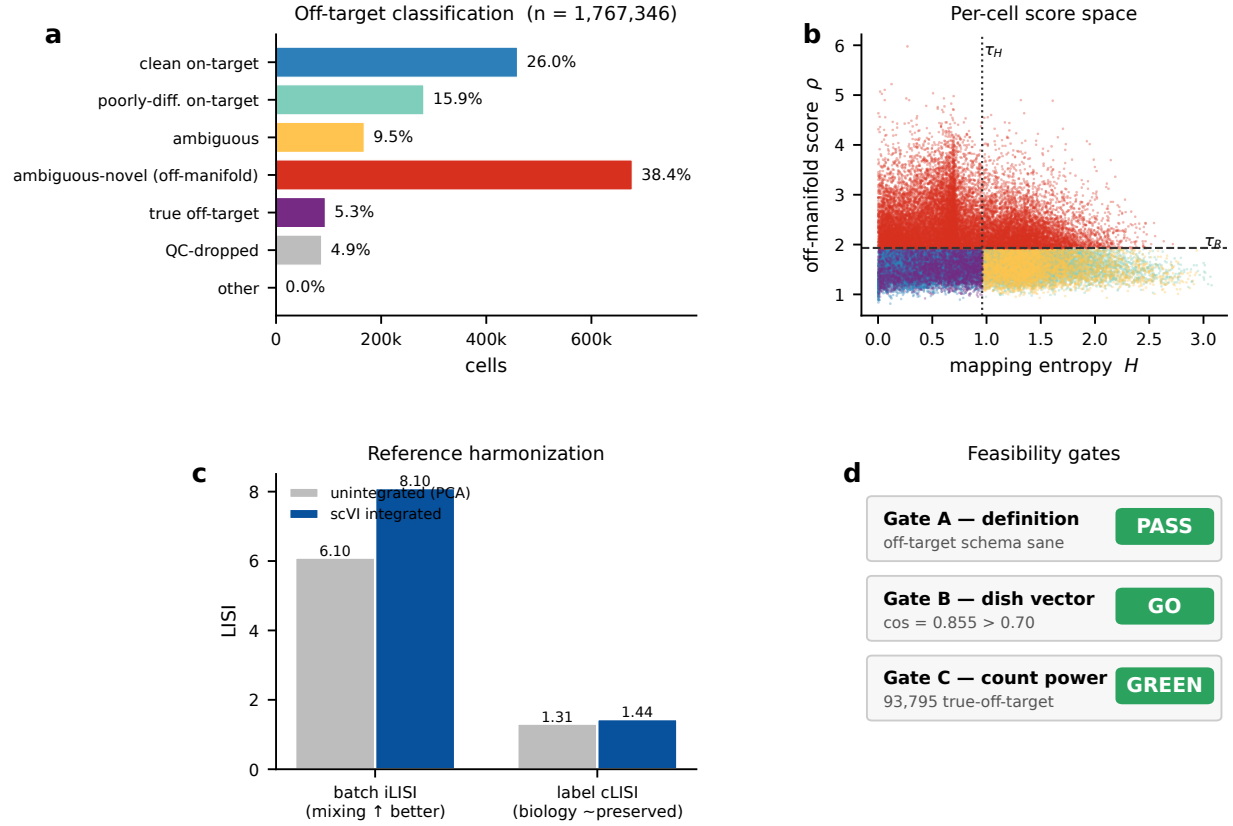


Figure 2. Harmonized mapping and the off-target landscape (Aim 1). (a) Six-state classification of all 1,767,346 organoid cells; the off-manifold *ambiguous-novel* state dominates (38.4%). (b) Per-cell score space: mapping entropy H versus off-manifold score ρ (subsample), coloured by class, with the calibrated thresholds τ_H (dotted) and τ_R (dashed). (c) Donor-batch integration improves mixing (iLISI 6.10 $\rightarrow$ 8.10) while preserving biology (cLISI 1.31 $\rightarrow$ 1.44). (d) The three feasibility gates pass: definition (A), dish-vector self-consistency (B, cosine 0.855), and off-target power (C, green).

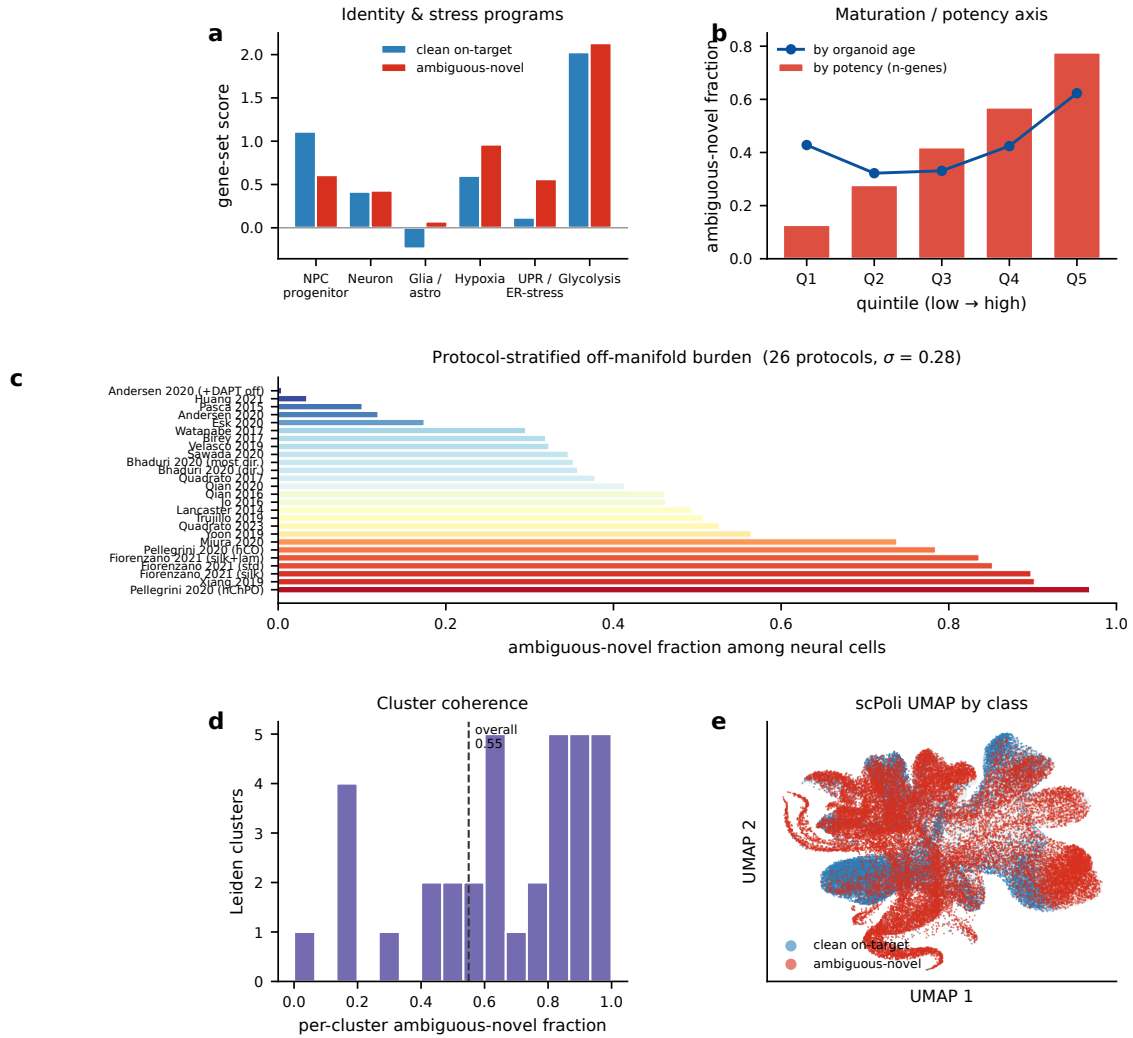


Figure 3. The off-manifold compartment is structured, not artifactual. (a) Ambiguous-novel neural cells lose neural-progenitor identity and gain hypoxia and unfolded-protein/ER-stress signatures relative to clean on-target neural cells. (b) Off-manifold fraction increases monotonically with a detected-gene potency proxy (bars) and with organoid age (line). (c) Protocol-stratified off-manifold burden spans 0.4%–96.8% across 26 protocols (colour = fraction); guided cortical protocols are lowest, choroid-plexus/midbrain/thalamic highest. (d) Per-cluster off-manifold fraction across 30 Leiden communities is bimodal (coherent populations, not scatter). (e) scPoli UMAP coloured by class: ambiguous-novel cells occupy contiguous territory.

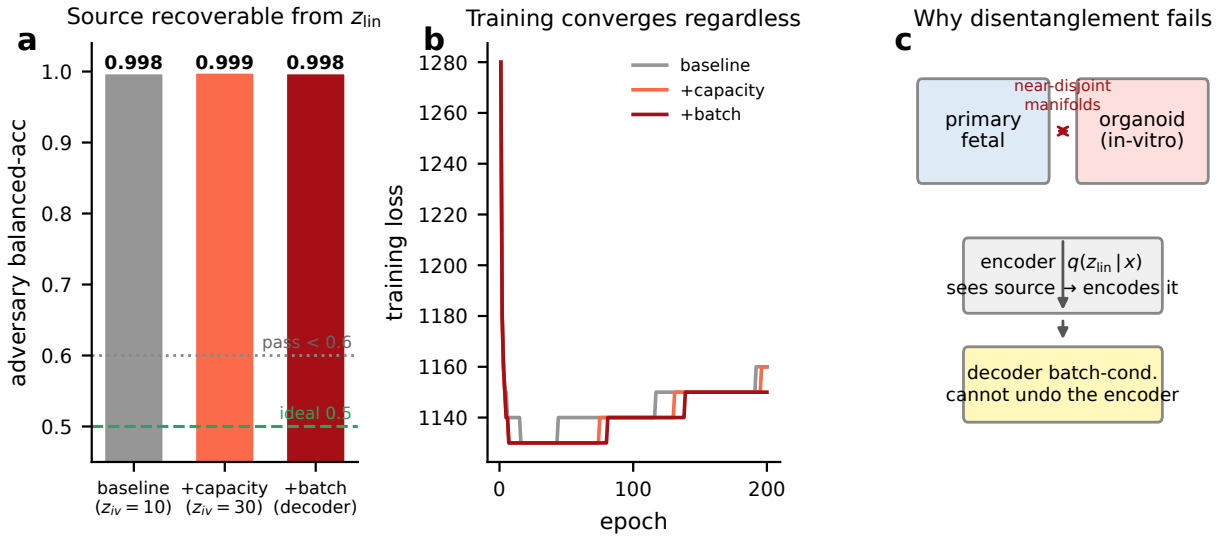


Figure 4. Lineage–dish disentanglement is not attainable (characterized negative; Aim 2). (a) An origin-decoding adversary recovers organoid-versus-primary identity from the “dish-stripped” latent z_{lin} at ≥ 0.998 balanced accuracy across all interventions (ideal 0.5; pass < 0.6). (b) Training converges normally in every run, so the failure is structural, not optimization-related. (c) Mechanism: near-disjoint primary/organoid manifolds make the background *encoder* encode source directly; decoder batch-conditioning cannot remove it.

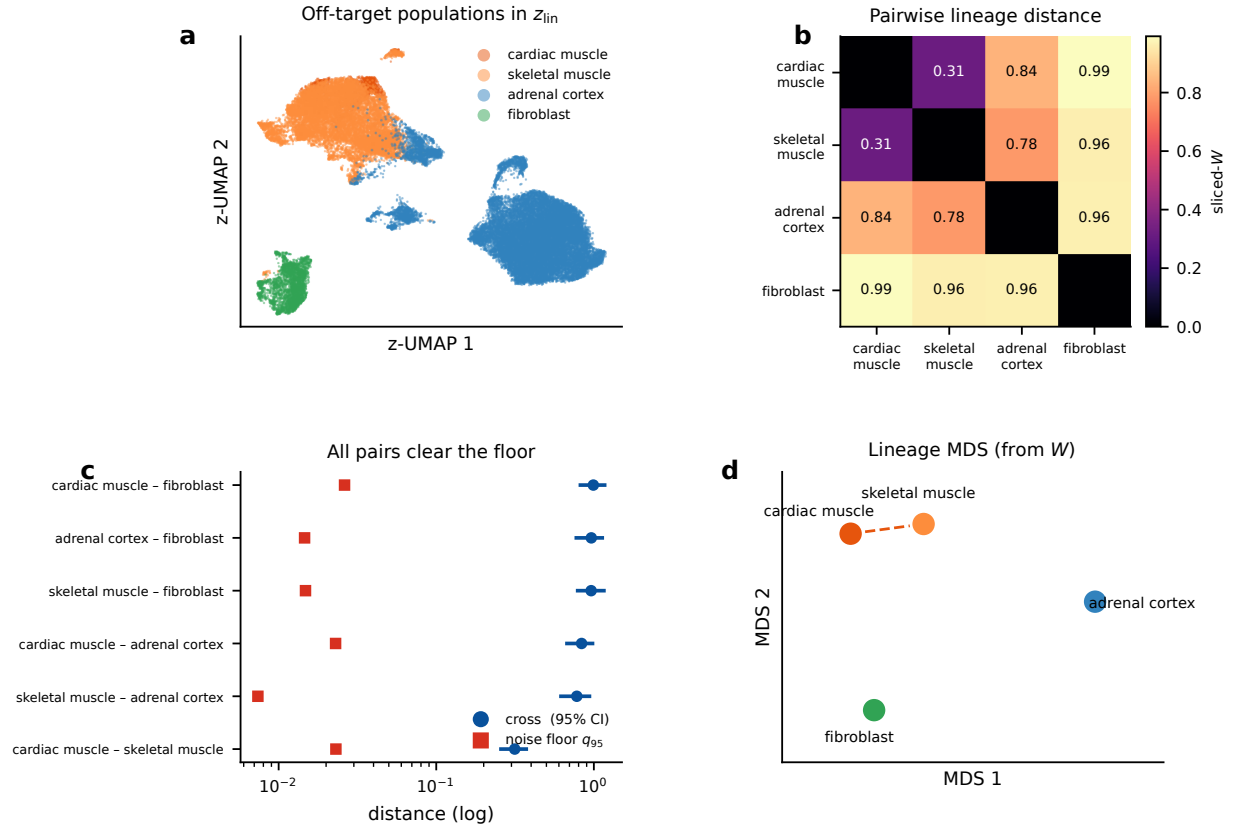


Figure 5. Off-target geometry is real and biologically ordered (Aim 3). (a) Background-latent (z_{lin}) embedding of the off-target populations; the four well-powered lineages separate. (b) Pairwise sliced-Wasserstein lineage-distance matrix. (c) Every well-powered pair clears its matched- N noise floor (q_{95} , red) by 14–105 $\times$ (cross-distance with 95% CI, blue; log axis). (d) Multidimensional scaling of the distance matrix: the two myogenic populations (cardiac, skeletal muscle) are closest, fibroblast most distant, as expected for a lineage embedding.

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